臺師大翻譯所 推廣班 115-2【筆譯6小時工作坊】AI翻譯科技入門

The Agentic AI Translator's Workbench

Moving from Chatbot Sessions to Productive Workflows

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About the instructor
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- 自2008起擔任自由譯者：財經、大學法規、ESG報告、新聞、歷史
- 臺大翻譯碩士學位學程兼任講師（2022起）
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本課程電子資源

https://github.com/ruben-tsui

- ➔ 點選左上角 ***Repositories***
- ➔ 點選 **NTNU_GITI_TT_261101**

Agenda

1. **Environment Setup**: Getting started with **Google Cloud Shell**.
2. **Task 1: Bitext Alignment**: Aligning a Taiwan Panorama article pair at the sentence level; extracting glossary entries
   《光華雜誌》文本對齊: 句對齊；術語及專有名詞擷取
3. **Task 2: Wuxia Translation**: Instructing AI to translate proper names and martial arts terminology consistently in a wuxia novel with a curated glossary 正確翻譯專有名詞及術語
4. **Task 3: OCR**: Extracting text from scanned Chinese text
   以多模態 AI 模型進行掃描文件文字辨識

Our Computing Environment for the Day

Cloud-based; No Installation Required

Google Cloud Shell

1. **Open your browser** and go to: `https://shell.cloud.google.com`
2. **Sign in** with your Google Account
3. **Click "Authorize"** if prompted to start the terminal
4. We'll cover the **Editor** and **Terminal**
5. What it is: A cloud-based Linux computing environment

Introducing the AI arness

Antigravity CLI

練習資料包下載網址

https://github.com/ruben-tsui

- ➔ 點選左上角 *Repositories*
- ➔ 點選 NTNU_GITI_TT_261101

You can install Antigravity CLI on your own computer with the command:

Visit the official website for more info:

https://antigravity.google/product/antigravity-cli

What is Antigravity CLI?

(CLI = command-line interface)
(opposite of GUI = graphical user interface)

- **Anthropic** released **Claude Code** in Feb. 2025 as an “agentic” command line tool that enables developers to delegate coding tasks directly from their terminal.
- Other AI companies soon followed suit:
  - **Google**: **Gemini CLI** (June 2025) ➔ **Antigravity** (Nov. 2025) ➔ **Antigravity CLI** (August 2026)
  - **OpenAI**: **Codex** (May 2025; GUI, not CLI); now **ChatGPT** app
  - Open-source projects: **OpenCode**; **OpenClaw** (Nov. 2025)

In a nutshell, Antigravity CLI …

- … is a coding agent powered by Google's LLMs; it can generate computer programming code (e.g., Python)
- But it's more than that. You can instruct it to do much more.
- Evolution of AI Technology:
  - LLM (Generative AI): A language model that predicts the next likely token (word, audio segment, pixels, etc.)
  - Chatbot: Built upon an LLM to interact with humans and to answer questions
  - AI Agent (Agentic AI): Uses an LLM as its "brain" to **design** and **execute** a solution to solve a problem – AI **harness**

Task 1:
Bitext Alignment & Glossary Extraction (1/3)

- **Scenario: Taiwan Panorama** article from:
  - https://www.taiwan-panorama.com/Articles/Details?Guid=4183d164-899b-4f49-af34-21abc8eea9de
- **Goal**: Align the Chinese (source) and its English translation/adaptation at the sentence level, and harvest bilingual cultural terms
- **What to do**: Create a file **TWP.md** in the **Editor** with the following instructions (**.md** stands for **Markdown**)**:**

Task 1:
Bitext Alignment & Glossary Extraction (2/3)

Tasks

1. Perform bitext alignment at the sentence level on the following pair of files @tea.zh.txt (source) and @tea.en.txt (target). Use the LLM to perform the alignment instead of a Python script. Place each aligned segment pair on the same line, with a tab character separating the source and target segments. If there's no matching target sentence(s), just leave the target string empty. Name the aligned text "tea.zh-en.txt".

2. Extract a glossary of Taiwan-specific cultural terms. Include as many entries as possible, but no fewer than 30. Place each zh-en glossary pair on the same line, with the two parts separated by a tab character. Name the glossary file "tea.glossary.zh-en.txt".

Task 1:
Bitext Alignment & Glossary Extraction (3/3)

- Next launch Gemini CLU by typing **agy** (followed by ENTER) in the Cloud Shell Terminal. [`/model` to change models]
- Type the command at Antigravity CLI's own input prompt:
  - **Execute @TWP.md please.**

    Then hit ENTER; wait for results (the 2 generated output files)
- You may download the generated files to your laptop by right-clicking the file name and select "Download…".

Task 2:
Instruct AI to adopt your glossary when translating (1/2)

- **Scenario:** You'd like to translate a few chapters of Jin Yong's Legend of the Condor Heroes from Chinese into English.
- **Challenge**: Consistently translated proper names and Kung Fu terminology across chapters
- **The Solution**: Provide a glossary as a "source of truth" (ultimate authority that overrides the model's own "internal knowledge")
- **What to do**: Create a file `CONDOR.md` with the following instructions**:**

Task 2:
Instruct AI to adopt your your glossary when translating (2/2)

```
## Tasks

1. Translate the source text @Condor_Ch04.zh.txt, which is from
a modern Chinese martial art (wuxia) novel, into idiomatic
English, using the glossary of proper names and special terms
from the file @Condor_glossary.txt .

2. Name the English translation "Condor_Ch04.en.txt".
```

Exercise: Ask Antigravity CLI to align **Condor_Ch04.zh.txt** and **Condor_Ch04.en.txt** at the sentence level, and then spot check to see if the LLM has translated names and terms according to your glossary.

Task 3:
OCR'ing Classical Chinese Text

Scenario: You'd like to extract editable text from a scanned, vertically printed Classical Chinese text in PDF format via OCR (optical character recognition)

Challenge: Scanned PDFs of vertically written/printed text; text with tables

The Solution: Gemini 3 Flash, a multimodal (text + vision) LLM, AI that "sees" and understands context

The Prompt:

```
Extract Chinese text from @南宋抗金.pdf and save the output to a Markdown
file with the same file name and the file extension .md please. Use the
LLM's OCR capabilities and not a Python program to perform OCR. Ignore
page boundaries and numbers and reconstitute broken sentences and
paragraphs into complete units. Create Markdown tables if the PDF pages
contain tables. Place all footnotes in a separate section at the end of
the Markdown file, with each footnote on its separate line and add a blank
line between footnotes.
```

Cloud Shell Workflow Summary

- **Viewing files in Editor**: Click the “**Open Editor**” (Pencil Icon) in the top-right toolbar.
- **Accessing the Terminal**: Click the “**Open Terminal**” (screen Icon)
- **Uploading files**: Drag and drop files from your laptop into Cloud Shell editor’s Sidebar (File Area).
- **Downloading**: Right-click any file in the Editor Sidebar ➔ **Download...** to transfer results from the cloud to your laptop.
- **Closing**: Type `exit` in the terminal and close the web page when done

So, what else can you do with an AI agent?

- **Review translation quality** (your own, another translator's):
  - Grammar, vocabulary choice, register, style
  - Adherence to "official" term base or glossary
- **Collect bilingual texts** (news articles, laws, regulations) to build parallel corpora
- **Translate a new document** based on/in inference to past translations of documents of the same type. The challenge is to provide the right (relevant) context to steer translation output.
  - ➔ Remember, "glossary" is one type of "context".

Appendix: Common Bash Commands

- **`ls -l`**
  - list contents of current folder (`-l` means long listing with file details)
- **`cd`**
  - change directory (navigate to a folder in the filesystem tree)
- **`pwd`**
  - print working directory
- **`unzip`**
  - unzip a .zip archive
- Type Ctrl-L (both Windows an macOS) to clear the screen

Thank You! Time for Q&A, but first …

Remember to shut down: Type `exit` in your terminal now. Google provides only 50 hours of free Cloud Shell usage per week, so exit when you're done to prevent idleness from burning through your quota.

Bonus!!

全國法規資料庫 華英檢索系統

https://huggingface.co/spaces/rubentsui/TWL2

A bilingual concordancer app I "vibe-coded" in two days, from raw data collection, data preprocessing, and web UI design and coding.